INCH-POUND

FF-S-611B
9 October 2025
SUPERSEDING
FF-S-611A
26 June 1972

FEDERAL SPECIFICATION

SPIKES, TRACK, SQUARE-SHANK

The General Services Administration has authorized the use of this federal specification by all federal agencies.

1. Scope and Classification

1.1 Scope. This specification covers square-shank, cut-steel track spikes (see 6.1).

1.2 Classification.

1.2.1 Grades. Spikes may be of the following grades, as specified (see 6.2).

Grade A - High-carbon steel.

Grade B - Soft steel.

1.2.2 Sizes. Spikes may be of the sizes as specified (see Table IV and 6.2)

2. APPLICABLE DOCUMENTS

2.1 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on the date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org or the American Society for Quality, 611 East Wisconsin Avenue, Milwaukee, WI 53202.)

Comments, suggestions, or questions on this document should be addressed to DLA Troop Support – Construction and Equipment (C&E) (ATTN: Code FCSB), 700 Robbins Avenue, Philadelphia, PA 19111-5098 or email to CE_Standardization@dla.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 2250

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AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-A751 - Standard Test Methods and Practices for Chemical Analysis of Steel Product
ASTM-E1282 - Standard Guide for Specifying the Chemical Compositions and Selecting Sampling Practices and Quantitative Analysis Methods for Metals, Ores and Related Materials

(Copies of this document are available from www.astm.org or the American Society for Testing and Materials, 100 Barr Harbor Drive, W. Conshohocken, PA 19428-2959.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the reference cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. Spikes shall be of steel manufactured by the open-hearth, acid-bessemer, electric-furnace or basic-oxygen processes. The chemical composition shall be as specified in Table I.

3.1.1 Chemical analysis. A chemical analysis of the steel used in the spikes shall be furnished by the manufacturer and shall show the carbon content and the copper content when copper-bearing steel is specified.

TABLE I. Chemical Composition.

Material	Carbon Content % Min		Copper* Content % Min
	Acid-Bessemer	Other	
Grade A	.20	.30	.20
Grade B	.06	.12	.20

*When copper-bearing steel is specified (see 6.2).

3.2 Tensile properties. The manufacturer may, at his option, substitute tension tests for the chemical analysis specified in 3.1.1. Tensile properties of the spikes shall be as specified in Table II.

TABLE II. Tensile Properties.

Material	Tensile Strength PSI Min	Yield Point PSI Min	Elongation In 2 inches % Min
Grade A	70,000	35,000	25
Grade B	55,000	27,500	25

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3.3 Bending properties.

3.3.1 Grade A.

3.3.1.1 Stem (body). The body of a finished spike shall withstand being bent cold through 120° around a pin having a diameter not greater than the thickness of the spike, without cracking on the outside of the bent portion.

3.3.1.2 Head. The head of a finished spike shall withstand being bent backwards to an angle of 55° with the line of the face of the spike, without cracking on the outside bent portion.

3.3.2 Grade B.

3.3.2.1 Stem (body). The body of a finished spike shall withstand being bent cold through 180°, flat on itself, without cracking on the outside of the bent portion.

3.3.2.2 Head. The head of a finished spike shall withstand being bent backward to the line of the face of the spike, without cracking on the outside of the bent portion and without showing evidence of forging laps on the surface of the bent portion.

3.4 Dimensions. Head shape and dimensions for spikes 5 inches or less in length shall be as commercially manufactured, unless otherwise specified (see 6.2 and Table IV). Spikes more than 5 inches in length shall conform to figures 1, 2 and 3, as applicable. Tolerances shall be as specified in Table III. All dimensions are in inches, unless otherwise specified.

TABLE III. Tolerances.

Cross-section	+.031	-.016
Head	+.094	-.031
Length, under head to point	+.125	-.125
Angle, under side of head	+1 degree	-1 degree

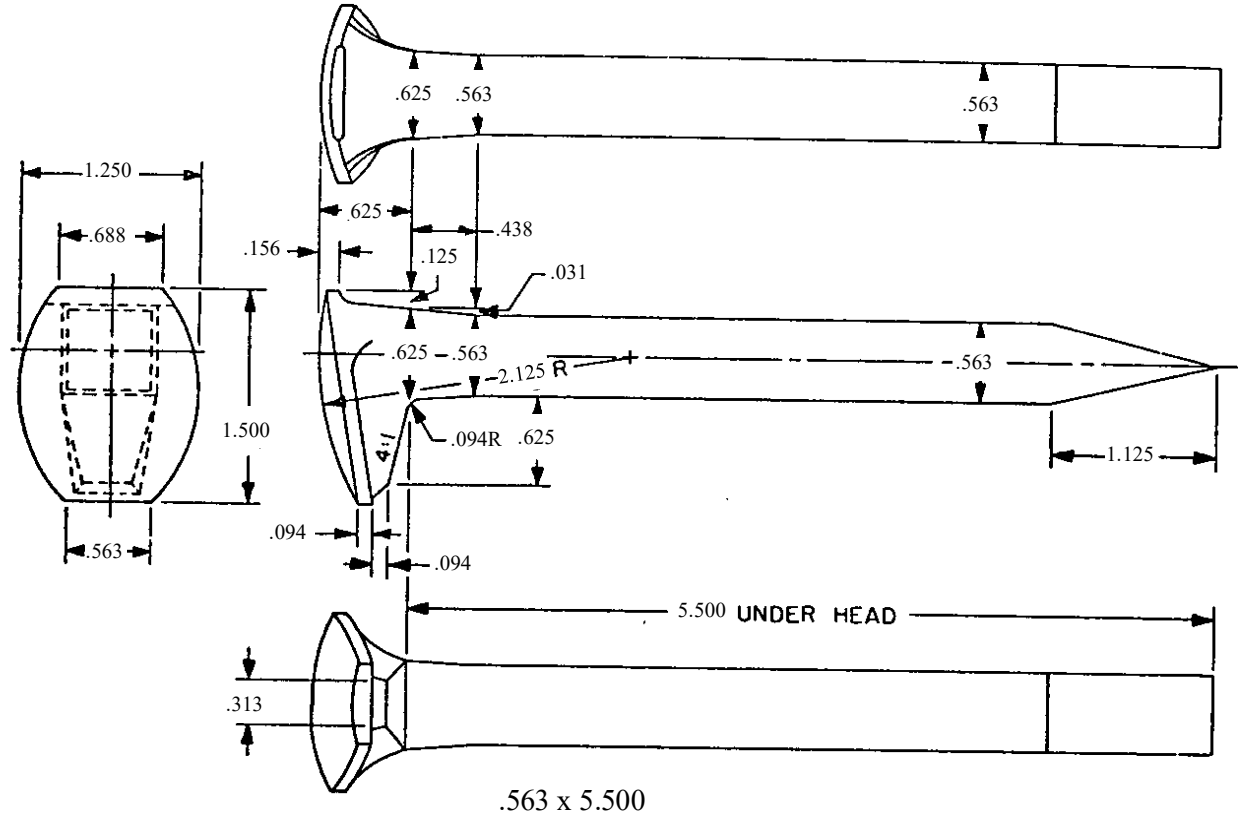
3.5 Marking. The heads of all spikes shall carry brand marks or letters so that the manufacturer may be readily determined.

3.5.1 High-carbon steel (grade A) spikes shall be marked with the letters “HC”.

3.5.2 Spikes specified to be of copper-bearing steel shall be marked with the letters “CU”.

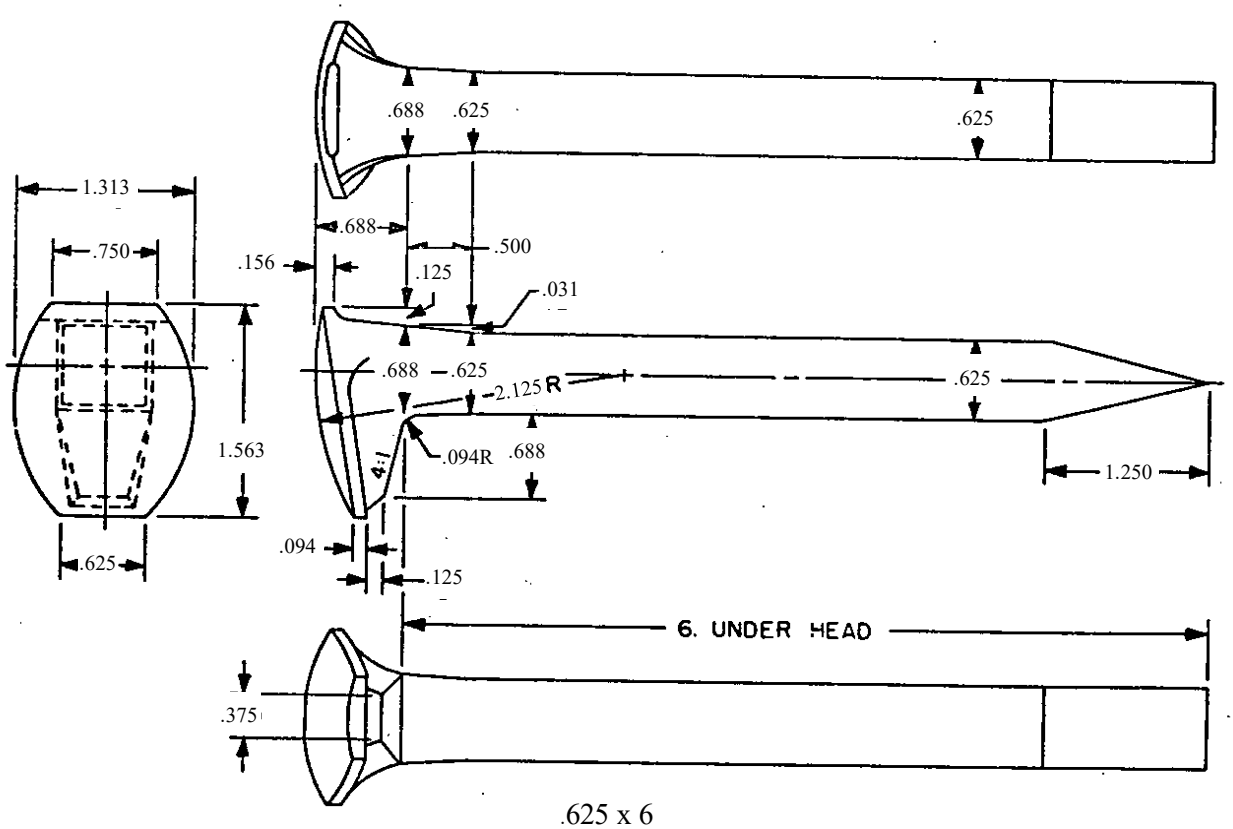
3.6 Workmanship. Finished spikes shall be smooth and straight with well-formed heads, sharp points, and be free from nicks, checks, cracks and ragged edges.

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.563 x 5.500

FIGURE 1.

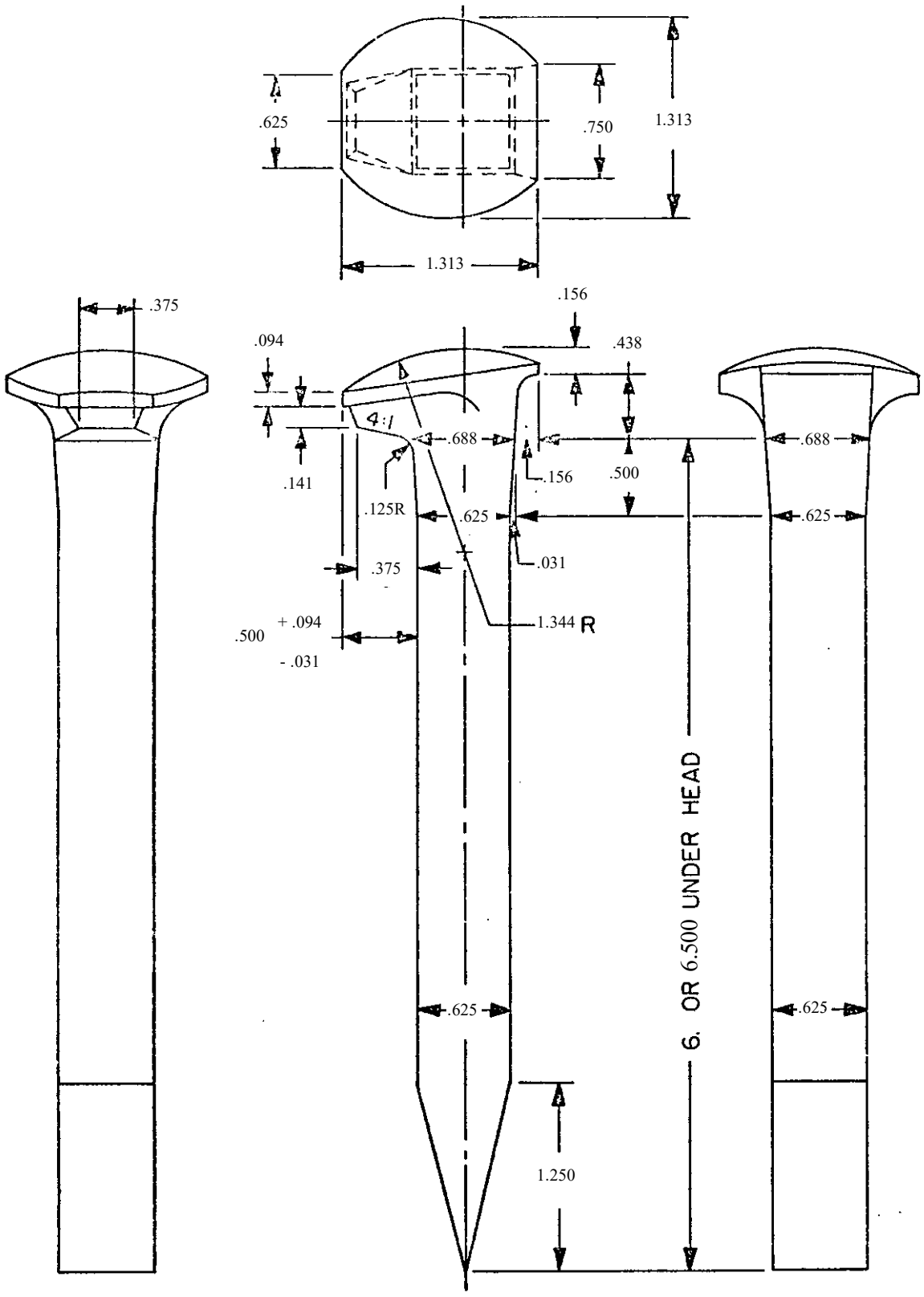


.625 x 6

(FOR OPTIONAL DESIGN SEE FIGURE 3)

FIGURE 2.

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.625 x 6

(OPTION FOR FIGURE 2.)

.625 x 6.500

FIGURE 3.

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TABLE IV. Sizes and Weights.

Size		Nominal Number of Spikes Per 200 Pounds	Size		Nominal Number of Spikes Per 200 Pounds
Cross Section	Length Under Head		Cross Section	Length Under Head	
.250	1.250	6,800	.375	5	885
.250	1.500	5,735	.438	2.500	1,105
.250	2	4,350	.438	3	1,055
.250	2.250	3,850	.438	3.500	875
.250	2.500	3,500	.438	4	775
.250	3	3,070	.438	4.500	710
.313	1.500	3,440	.438	5	650
.313	2	2,875	.500	2.500	890
.313	2.250	2,760	.500	3	780
.313	2.500	2,460	.500	3.500	690
.313	3	2,165	.500	4	595
.313	3.500	1,970	.500	4.500	545
.313	4	1,775	.500	5	515
.375	2	1,880	.563**	4.500	410
.375	2.250	1,720	.563**	5	365
.375	2.500	1,560	.563	5.500	*335
.375	3	1,365	.625**	4.500	310
.375	3.500	1,210	.625**	5	285
.375	4	1,080	.625	6	*245
.375	4.500	975	.625	6.500	*230

NOTE: The number of finished spikes varies as much as ± 15%, Except for those marked with an (*), which vary about ± 5%.

** Heads are similar to the designs shown in figures 1 and 2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his/her own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Lot. Unless otherwise specified (see 6.2), a lot shall consist of not more than 10 tons of spikes of the same heat and size manufactured by the same process and offered for acceptance at any one time. When specified (see 6.2), a number of lots may be offered simultaneously for acceptance.

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4.3 Sampling.

4.3.1 Sampling for examination. A random sample of spikes shall be taken from each lot in accordance with ASQ-Z1.4, Inspection Level S-1. The Acceptable Quality Level (AQL) shall be 6.5 percent defective.

4.3.2 Sampling for chemical analysis.

4.3.2.1 Melt known. Samples shall consist of one test ingot cast during the pouring of each heat of steel.

4.3.2.2 Melt unknown. Sample drillings shall be taken from three sample spikes selected at random from each 10 ton lot of spikes offered for acceptance.

4.3.3 Sampling for tension test. When the manufacturer requests to conduct a tension test instead of chemical analysis, one sample spike shall be selected from each 10 ton lot or fraction thereof for the tension test.

4.3.4 Sampling for bend test. One sample spike shall be selected from 5 tons or fraction thereof in a lot for the stem (body) bend test and another sample spike for the head bend test.

4.4 Examination. Samples selected as specified in 4.3.1 shall be subjected to visual inspection for marking (3.5) and workmanship (3.6), and to measurements (using standard inspection equipment) for dimensional requirements (3.4).

4.5 Tests.

4.5.1 Chemical analysis. Samples selected as specified in 4.3.2 shall be subjected to a chemical analysis to verify conformance to 3.1.1 and Table I. The analysis shall be conducted in accordance with ASTM-E1282 or ASTM-A751. In case of dispute, ASTM-E1282 shall be the basis for acceptance.

4.5.2 Tension test. When requested by the manufacturer, samples selected as specified in 4.3.3 shall be subjected to a tension test to verify conformance to 3.2 and Table II.

4.5.3 Bend test. The bend test may be made by pressure or hammer blows.

4.5.3.1 Stem (body). Samples selected as specified in 4.3.4 shall be subjected to a bend test to verify conformance with 3.3.1.1 and 3.3.2.1, as applicable.

4.5.3.2 Head. Samples selected as specified in 4.3.4 shall be subjected to a bend test to verify conformance with 3.3.1.2 and 3.3.2.2, as applicable.

4.6 Rejection, re-examination and retest.

4.6.1 Rejection. Failure of any sample to meet the chemical or mechanical requirements of this specification shall be cause for the rejection of the lot.

4.6.2 Re-examination. A lot rejected for failing to meet the marking, workmanship or dimensional requirements may be resubmitted for examination provided the manufacturer has examined all the spikes in the lot for the defect noted and has removed and replaced, reworked or repaired all nonconforming spikes.

4.6.3 Retest. A lot rejected for failing to meet any test requirements may be resubmitted for retest in accordance with the requirements of ASQ-Z1.4.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of material is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that is helpful, but is not mandatory.)

6.1 Intended use. The larger spikes covered by this specification are intended to hold down railroad rails to the tie and tie plate. Smaller spikes are intended for general purpose use wherever an offset head is required.

6.2 Ordering data. Purchasers should exercise any desired options offered herein, and procurement documents should specify the following:

a. Title, number, and date of this specification.
b. Grade, and if copper-bearing steel is required. If grade is not specified, either grade may be furnished (see 1.2.1).
c. Size: cross-section and length (see 1.2.2 and Table IV).
d. Dimensions, if other than specified (see 3.4).
e. Lot size and number, if other than specified (see 4.2).
f. Packaging requirements (see 5.1).

6.3 Subject term (key word) listing.

Cut-steel
High-carbon steel
Soft steel

6.4 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodians:
Army - AR
Navy - YD

Preparing activity:
DLA - CQ

(Project 2250-2025-001)

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